Gradient Descent

Trust the line, one small step at a time

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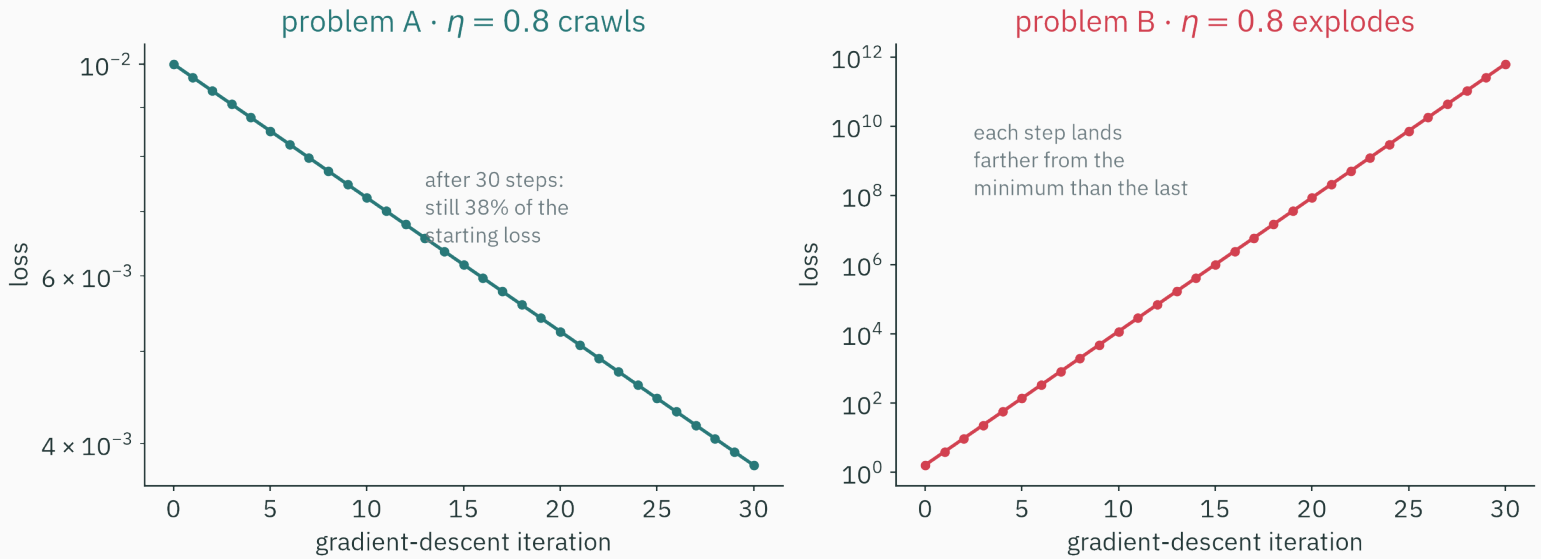
Mathematical Foundations for AI · IIT Gandhinagar

Two quadratic objectives, one learning rate

The same learning rate can converge or diverge

V VISUAL

Gradient descent uses the same code and learning rate $\eta = 0.8$ on two quadratic objectives:



It converges slowly on one and diverges on the other. This lecture explains both outcomes with exact calculations.

Why one learning rate behaves differently

Start by testing whether $\eta = 0.8$ is simply too large or too small:

Possible explanation	**What it would predict**	**Observation**
η is too **big**	both problems overshoot and explode	✗ A crawls
η is too **small**	both problems crawl	✗ B explodes

The same number cannot be simultaneously too big and too small…

…unless "big" and "small" are measured relative to something in the problem. Naming that something is today's whole job.

Everything we need is already on the table

from	what it gave us
L7	the local line: $f(x_0 + h) \approx f(x_0) + f'(x_0)h$ — "L17 will quote today's line"
L8	the direction: $-\nabla f$ is steepest descent (Cauchy–Schwarz) — we even ran a small GD
L16	the license: on convex bowls, local information is globally trustworthy

- L8 settled **which way** to step. Nobody has yet said **how far**.
- Today: derive the algorithm, then analyze how η interacts with curvature.

Learning outcomes

By the end of this lecture you will be able to:

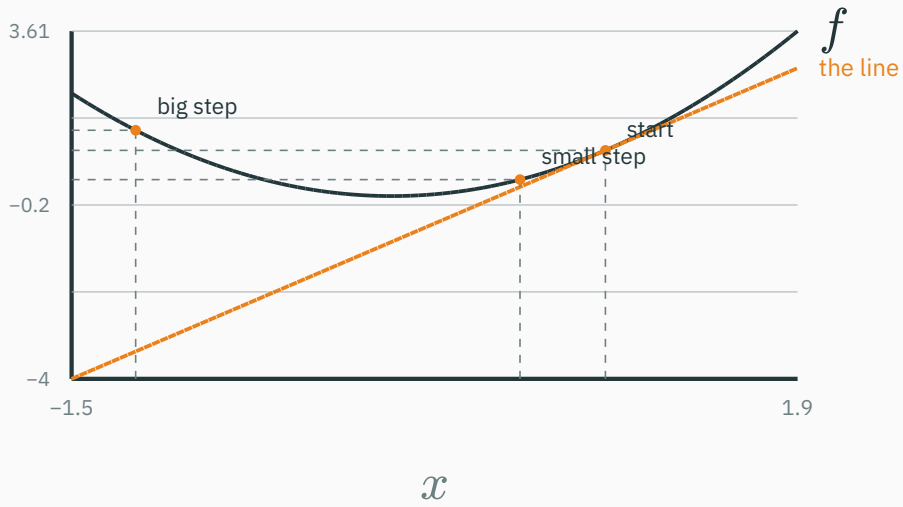
1. **Derive** gradient descent from the first-order Taylor approximation developed in L7.
2. Implement the **four-line algorithm** and read η as a trust radius.
3. Predict **crawl / converge / oscillate / diverge** from η and the curvature — the wall at $\eta = 2/a$.
4. Run the **exact analysis on quadratics**: contraction factor $1 - \eta a$, per-axis rates.
5. Compute a **condition number** κ and explain the zig-zag pathology it causes.
6. Explain why **feature scaling** speeds up training, and how **momentum** fights ravines.

Deriving gradient descent from Taylor's approximation

The first-order Taylor model

V VISUAL

At $x = 1$ on $f(x) = x^2$, the derivative defines the local linear approximation.



A small step decreases the true function. A large step can increase it even though the linear approximation continues downward. The approximation is local.

The approximation is accurate only for small steps

L7′s local replacement, in today's many-variable clothes (L9): for a step Δ from θ,

$$f(\theta + \Delta) \approx f(\theta) + \nabla f(\theta)^\top \Delta$$

- The error of this replacement grows like $\|\Delta\|^2$ (the second-order Taylor term) — **double** your step, the line lies **four times** harder.
- The linear model is useful only over a neighbourhood in which the neglected curvature term remains small.

This is the spine of the lecture: **trust the linear model for one small step, then rebuild the line and repeat**. Everything else is bookkeeping about r.

Which direction? L8 already answered

Inside the trust radius, pick the step the **line** likes best:

$$\min_{\|\Delta\| \le r} \; \nabla f(\theta)^\top \Delta$$

L8 callback — by Cauchy–Schwarz the dot product is most negative when Δ points **opposite** the gradient:

$$\Delta^\star = -r \, \frac{\nabla f(\theta)}{\|\nabla f(\theta)\|}$$

- Direction: $-\nabla f$, dead against the gradient — **steepest descent**, exactly L8's compass.
- New today: the length r — the part L8 left open.

The update rule

Fold the radius and the gradient's length into one knob, $\eta > 0$:

$$\theta_{t+1} = \theta_t - \eta \nabla f(\theta_t)$$

- η is the **learning rate**: it scales the gradient, so the step length is $\eta \|\nabla f(\theta_t)\|$.
- Rebuild the line at the new point, step again. That is the entire method.

This is the application anticipated in L7: the update uses today's local linear approximation to choose the next point.

A sufficiently small step decreases the objective

For differentiable f, substitute $\Delta = -\eta \nabla f$ into the first-order expansion:

$$f(\theta - \eta \nabla f) = f(\theta) - \eta \|\nabla f(\theta)\|^2 + o(\eta)$$

- If $\nabla f(\theta) \neq \mathbf{0}$, the negative linear term dominates for sufficiently small positive η, so the true objective decreases.
- Near a stationary point $\nabla f \to 0$, so the step norm $\eta \|\nabla f\|$ shrinks.

The decrease above is a local first-order prediction. A step that is too large can increase the true objective, as in the previous figure.

The whole algorithm is four lines

```
theta = theta0                      # 1. start somewhere
for t in range(T):                  # 2. repeat
    g = grad(L, theta)              # 3. which way is uphill  (L8)
    theta = theta - lr * g          # 4. small trusted step downhill
```

- Stop when $\|g\|$ is tiny, the loss plateaus, or the budget T runs out. A small gradient alone does not distinguish a minimum from a saddle or plateau.
- This loop and its stochastic variants train many models, from linear regression to neural networks.

One hyperparameter, no memory, only first derivatives. The rest of today: what that one hyperparameter does.

Numbers first: halving your way down

$f(x) = x^2$ (curvature $f'' = a = 2$), learning rate $\eta = 0.25$, start $x_0 = 2$:

$$x_1 = 2 - 0.25 \cdot (2 \cdot 2) = 1, \quad x_2 = 1 - 0.25 \cdot 2 = 0.5, \quad x_3 = 0.25, \quad \dots$$

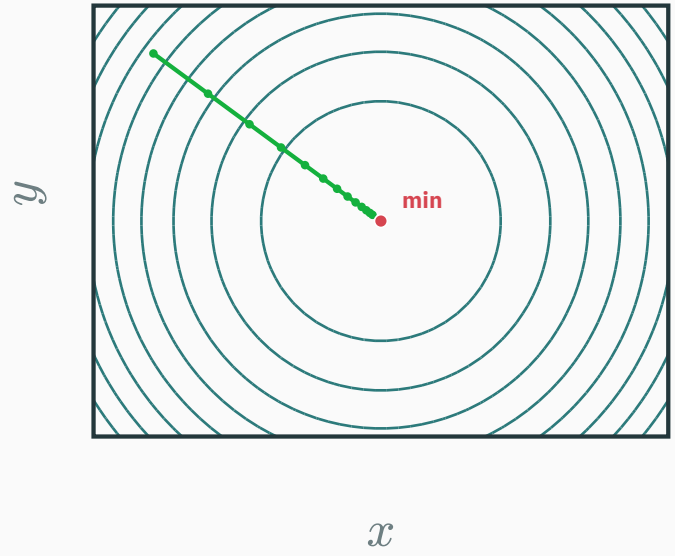
- Every step lands exactly **halfway** to the minimum: $x_{t+1} = x_t - 0.25 \cdot 2x_t = 0.5\, x_t$.
- The distance to the minimum shrinks by the **same factor 0.5 every step** — a clean geometric decay.

The next section derives this per-step factor exactly.

On a round bowl, GD is unstoppable

V VISUAL

$f(x, y) = x^2 + y^2$ — same curvature in every direction:



round contours $\Rightarrow -\nabla f$ points **straight at the minimum**; steps shrink as $\|\nabla f\| \to 0$

This round bowl is the well-conditioned reference case.

What η really is

- **Too small**: the guaranteed first-order decrease per step is much smaller than it could be.
- **Too large**: the neglected curvature dominates, and the true f may rise.
- **Just right**: the largest step the line can still honor.

How large is that, exactly? It must depend on how fast the line goes stale — the **curvature**. That is a number we can compute. Next section: compute it.

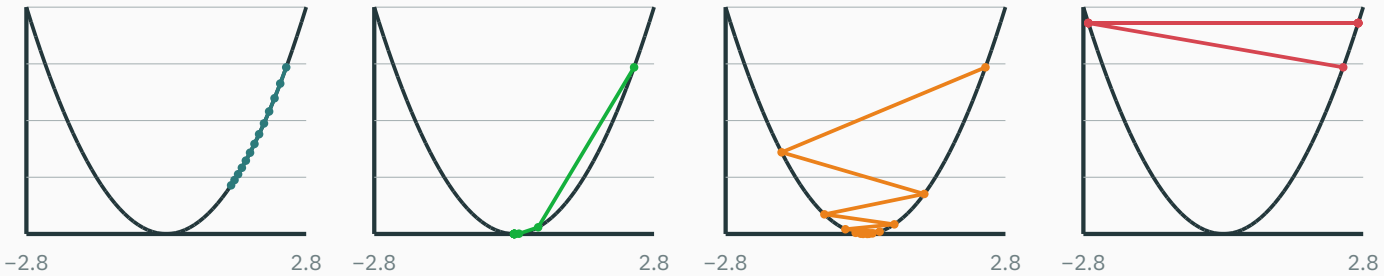
Choosing the learning rate

Four behaviours as η changes

V VISUAL

GD on $f(x) = \frac{1}{2}x^2$ (curvature $a = 1$), $x_0 = 2.4$, twelve steps — four choices of η:

$\eta = 0.05$ · **crawls** $\eta = 0.8$ · **converges** $\eta = 1.7$ · **oscillates in** $\eta = 2.15$ · **diverges**

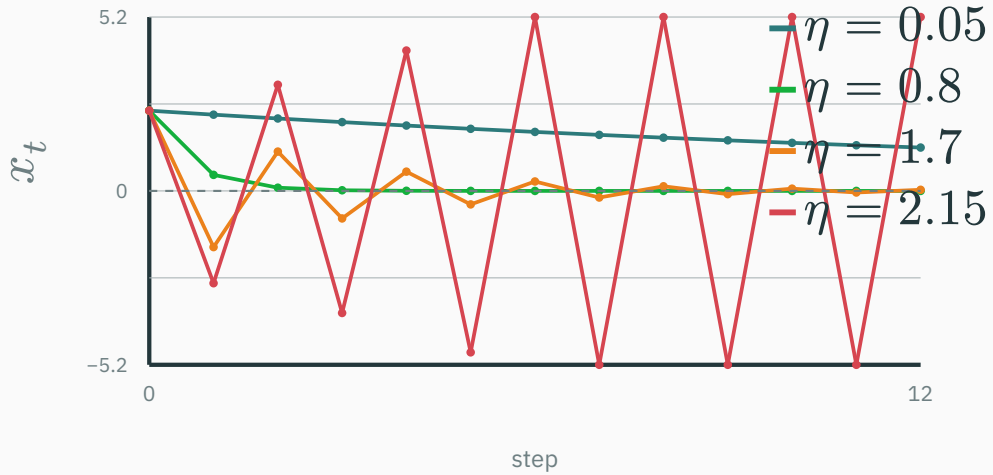


every dot is a real iterate, computed by this slide — crawl · converge · overshoot-but-converge · explode

The same four walks, in time

V VISUAL

Plot the iterate x_t itself against the step count:



- crawl and converge stay on one side; overshoot **crosses the minimum every step** and still converges; diverge crosses and **grows** (clipped at ± 5.2).

The wall sits at exactly $\eta = 2/a$

On a bowl of curvature a, the step from x has length $\eta a\ |x|$ — against a distance-to-minimum of $|x|$:

- $\eta a < 1$: land **short** of the minimum — same side, monotone approach.
- $1 < \eta a < 2$: land **past** the minimum but **closer** than you started — the overshooting dance.
- $\eta a > 2$: land past it and **farther out** — each step amplifies the error. Divergence.

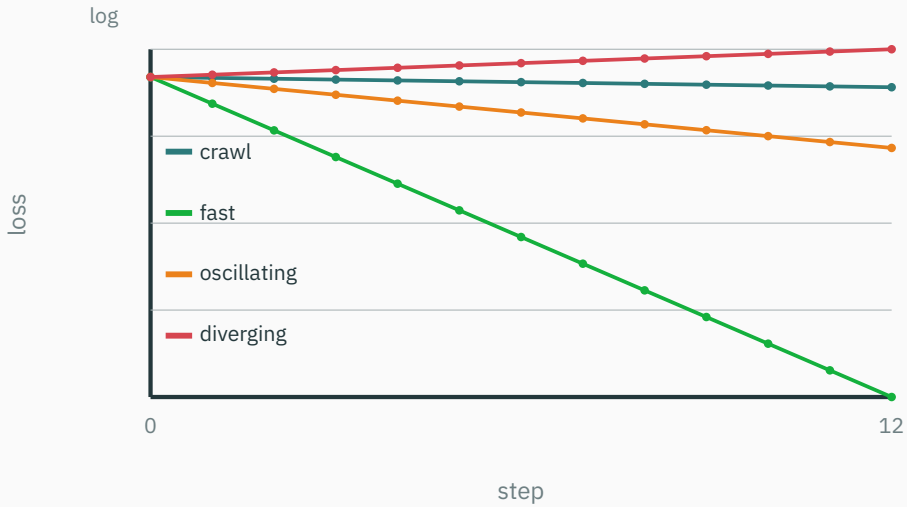
The speed limit is $\eta < 2/a$ — two divided by the curvature.

If f has an L-Lipschitz gradient, then $0 < \eta < 2/L$ guarantees a decrease at every non-stationary iterate. Convergence needs additional conditions such as a lower-bounded objective; convexity gives stronger conclusions.

What you'll actually see: the training curve

V VISUAL

In 10^6 dimensions you can't watch x_t — only the **loss**. Same four runs, log scale (L2 taught us to love logs):



geometric decay = **straight line** on a log axis; the slope is the per-step factor — divergence slopes **up**

You will diagnose learning rates from curves like these for the rest of your career.

Interactive: vary the learning rate

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/optimizer-race

Drop a start point on a 2-D loss surface and sweep the learning rate live: watch the same four personalities — crawl, converge, overshoot, explode — appear as you cross the $2/a$ wall.

Find the wall experimentally on the bowl, then check it against the curvature readout.

Checkpoint: the knife edge

Q QUESTION

Run GD on $f(x) = (x-3)^2$ — curvature $a = f'' = 2$ — with $\eta = 1$, from $x_0 = 0$. What happens?

A. Converges to $x = 3$ in one step

B. Diverges to infinity

C. Bounces between 0 and 6 forever

D. Converges slowly, like 0.5^t

Answer: the knife edge

A ANSWER

Correct: C — bounces between 0 and 6 forever

Why: The update is $x \leftarrow x - 2(x-3) = 6 - x$: each step **reflects** through the minimum. $\eta = 1$ is exactly the wall $2/a = 1$ — the error factor is -1: never shrinks, never grows.

$x:\quad 0 \to 6 \to 0 \to 6 \to \cdots$ — a perpetual, loss-preserving dance around $x^* = 3$

Exact analysis on quadratics

Why quadratics give a local model

Near a minimum θ^* of a smooth loss, use the Taylor model from L9:

$$f(\theta) \approx f(\theta^*) + \frac{1}{2}\, f''(\theta^*)(\theta - \theta^*)^2$$

- Near a non-degenerate smooth local minimum, the second-order Taylor model is approximately quadratic, with curvature $a = f''(\theta^*)$ in one dimension.
- And on a quadratic we don't have to approximate anything: GD can be solved **exactly**, in one line.

Exact answers on positive-definite quadratics describe local gradient-descent behavior near a smooth strict minimum.

⭐ One line of algebra

D DERIVATION

Take $f(x) = \frac{a}{2}\, x^2$ with curvature $a > 0$, so $f'(x) = ax$. Apply the update:

$$x_{t+1} = x_t - \eta\, a x_t = \underbrace{(1 - \eta a)}_{\text{one number}}\, x_t$$

Unroll it — the whole future in closed form:

$$x_t = (1 - \eta a)^t\, x_0$$

GD on a quadratic is repeated multiplication by the constant $1 - \eta a$.
Nothing else happens.

Check against the halving demo: $a = 2$, $\eta = 0.25 \Rightarrow 1 - \eta a = 0.5$. ✓

⭐ The contraction factor decides everything

D DERIVATION

Give $c = 1 - \eta a$ its name: the **contraction factor**. Error after t steps: $|c|^t \ |x_0|$; loss shrinks by c^2 per step.

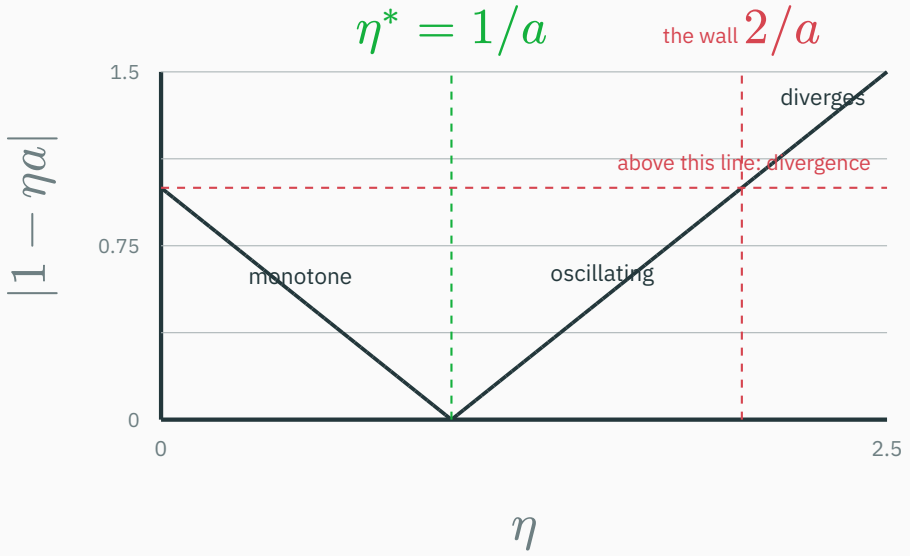
η range	factor c	behavior
$0 < \eta < 1/a$	$0 < c < 1$	monotone convergence (crawl if $c \approx 1$)
$\eta = 1/a$	$c = 0$	the minimum in **one step**
$1/a < \eta < 2/a$	$-1 < c < 0$	overshoots each step, still converges
$\eta = 2/a$	$c = -1$	bounces forever (the knife edge)
$\eta > 2/a$	$c < -1$	diverges

Convergence $\Leftrightarrow |1 - \eta a| < 1 \Leftrightarrow 0 < \eta < 2/a$**. The best** η **is** $1/a$ **— one over the curvature.**

The factor, as a picture

V VISUAL

$|1 - \eta a|$ against η (drawn for $a = 1$):



a V with its tip at $\eta = 1/a$: undershoot on the left branch, overshoot on the right, cross $|c| = 1$ and you burn

Apply the contraction factor to both objectives

D DERIVATION

The opening two objectives were $f(x) = \frac{a}{2}x^2$ with different curvatures and the same $\eta = 0.8$:

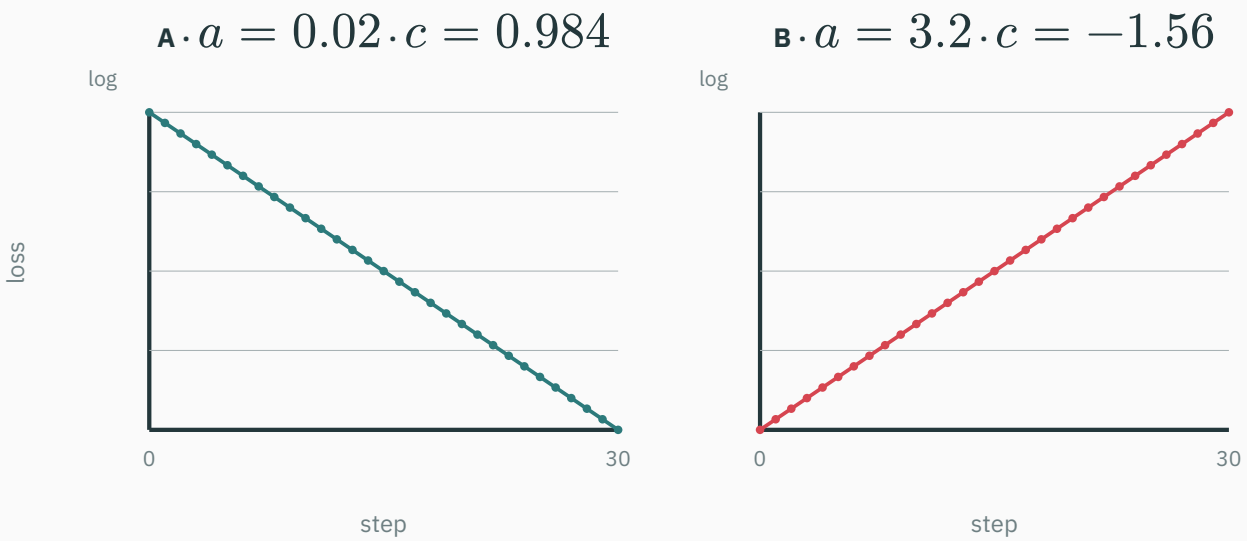
	problem A $\cdot\ a = 0.02$	**problem B** $\cdot\ a = 3.2$
factor $c = 1 - 0.8a$	0.984	-1.56
$\lvert c \rvert$ vs 1	just below — barely contracts	above — **expands**
after 30 steps	$0.984^{30} = 0.62$ of the error left $\Longrightarrow$ loss $\times\ 0.38$	error $\times\ 1.56^{30}$; loss $\times\ 2.43$ per step $\approx 10^{11}$

- L1′s figure annotated problem A with “still 38% of the starting loss” — that number is 0.984^{60}. **Predicted by one line of algebra.**
- For problem B, $\eta = 0.8$ sits beyond its wall $2/3.2 = 0.625$. It never had a chance.

Compute both trajectories

V VISUAL

The L1 figure again — except now **this slide computes it** from $x_{t+1} = (1 - \eta a)x_t$:



straight lines on the log axis — geometric decay and geometric explosion, slopes $2 \log|c|$

The same η is small relative to curvature 0.02 and too large relative to curvature 3.2.

⭐ One learning rate must serve every eigendirection

D DERIVATION

The deeper trouble: one problem can contain **both curvatures at once**. Take the diagonal bowl

$$f(x,y) = \frac{1}{2}(ax^2 + by^2), \quad \nabla f = (ax, by), \quad a \text{ flat}, \; b \text{ steep}$$

The update never mixes the coordinates — GD **decouples** into two 1-D stories with one shared η:

$$x_t = (1-\eta a)^t x_0, \qquad y_t = (1-\eta b)^t y_0$$

- Stability is hostage to the **steep** axis: need $\eta < 2/b$.
- Speed is hostage to the **flat** axis: its factor $1-\eta a$ then sits near 1.

⭐ One learning rate must serve every eigendirection

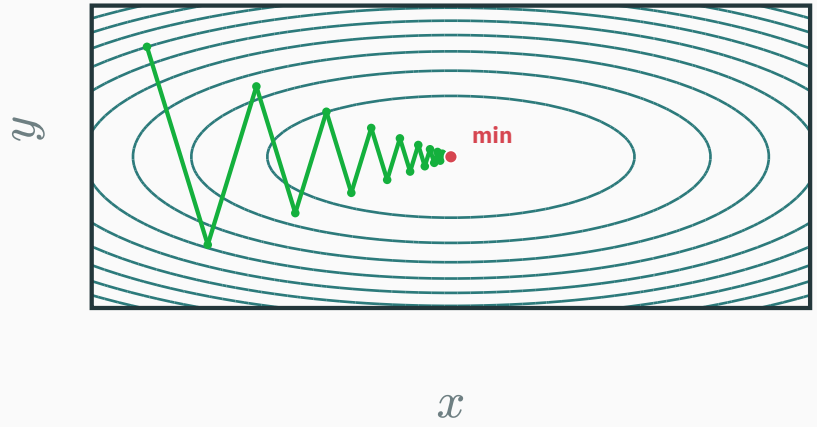
D DERIVATION

General bowls? L5 callback: eigenvectors of the Hessian rotate any bowl into this diagonal form — a and b are just its extreme eigenvalues. The diagonal case **is** the general case.

Watch both factors at once

V VISUAL

$f(x, y) = \frac{1}{2}(x^2 + 9y^2)$, $\eta = 0.2$: per-axis factors $1 - 0.2 \cdot 1 = 0.8$ and $1 - 0.2 \cdot 9 = -0.8$.



y flips sign every step (factor -0.8) while x shrinks steadily (factor $+0.8$) — both visible in one real trajectory

No η serves two curvatures

Try to pick one η for A's and B's curvatures **simultaneously** ($a = 0.02$, $b = 3.2$):

η	**flat axis ($a = 0.02$)**	**steep axis ($b = 3.2$)**
0.8 (L1′s)	$c = 0.984$ — crawls	$c = -1.56$ — explodes
$0.625 = 2/b$	$c = 0.9875$ — crawls	$c = -1$ — bounces forever
0.5	$c = 0.99$ — crawls **worse**	$c = -0.6$ — fine
$0.3125 = 1/b$	$c = 0.99375$ — crawls worse still	$c = 0$ — one step!
$50 = 1/a$	$c = 0$ — one step!	$c = -159$ — catastrophic

Every row sacrifices an axis. When curvatures are far apart, no learning rate is good — that situation has a name.

The condition number controls the convergence rate

D DERIVATION

$$\kappa = \frac{a_{\max}}{a_{\min}} \quad 0 < a_{\min} \le a_{\max} \text{ — for the opening SPD quadratic: } 3.2/0.02 = 160$$

Best compromise: $\eta^* = 2/(a_{\max} + a_{\min})$ balances the two factors at

$$|c| = \frac{\kappa - 1}{\kappa + 1} \approx 1 - \frac{2}{\kappa}$$

κ	1	10	160	10^4
best factor	0	0.818	0.9876	0.9998
steps to cut error $100\times$	1	≈ 23	≈ 369	$\approx 23,000$

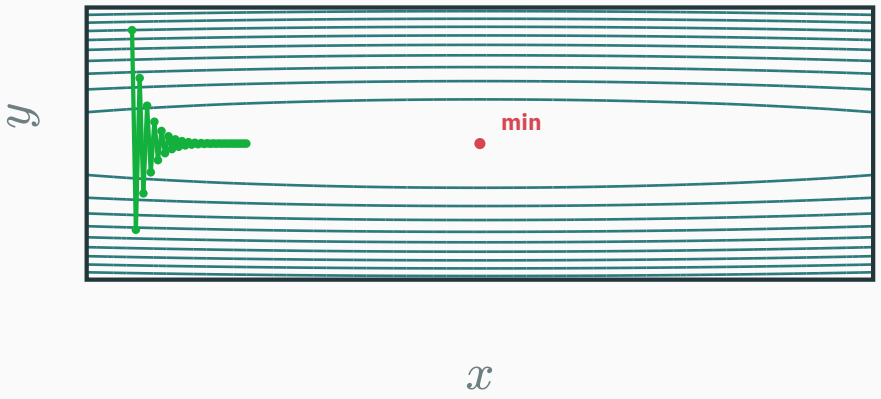
For this quadratic family, the number of gradient-descent steps grows in proportion to the condition number κ.

The ill-conditioned bowl

Both personalities in one problem

V VISUAL

Put the opening example's two curvatures into one objective: $f(x, y) = \frac{1}{2}(0.02x^2 + 3.2y^2)$, $\kappa = 160$, and run GD with a safe $\eta = 0.55$:



the infamous **zig-zag**: oscillate across the ravine (factor -0.76), crawl along it (factor 0.989)

In a local quadratic model, Hessian eigenvalues play the role of directional curvatures. A wide positive eigenvalue range produces the same slow/oscillatory tradeoff; zero or negative eigenvalues require separate analysis.

Why zig-zag? L8 told us

- L8 ⭐: every GD step is **perpendicular to the contour** it starts on.
- In a ravine, contours are stretched ellipses — perpendicular-to-the-contour points **across** the valley, hardly **along** it.
- The steep axis pins η below its wall $2/b$, so the along-valley axis moves at factor $1 - \eta a \approx 1 - 2/\kappa$: a crawl.

Zig-zag is not a bug in the code — it is steepest descent doing exactly what "steepest" means, on a bowl where steepest ≠ toward the minimum.

Where do squashed bowls come from? Your features

Fit a price from two features using the loss $\frac{\|X\boldsymbol{w}-\boldsymbol{y}\|^2}{2n}$. Its Hessian is $X^\top \frac{X}{n}$; the diagonal entry for weight w_j is $\text{mean}\left(x_j^2\right)$:

feature	**values**	**curvature** $\text{mean}\left(x_j^2\right)$
house area (m²)	$80, 120, 160$	$46400/3 \approx 15,467$
rooms	$2, 3, 4$	$29/3 \approx 9.67$

$$\text{diagonal scale ratio} = \frac{46400/3}{29/3} = 1600$$

Units can create a large coordinate-scale mismatch. Here there is a second problem: area $= 40 \times$ rooms, so the columns are perfectly dependent and the true Hessian condition number is infinite.

The fix costs one line: standardize

```
X = (X - X.mean(axis=0)) / X.std(axis=0)   # each column: mean 0, std 1
```

- Every column now has $\text{mean}(x_j^2) = 1$, removing conditioning caused purely by unequal units. Correlated columns can still make $X^\top X$ ill-conditioned, so standardization does **not** guarantee $\kappa \approx 1$.
- In this example the two standardized columns remain identical, so the Hessian is still singular. Standardization fixes scale, not collinearity.
- Standardization often improves gradient-based optimization by rescaling coordinate directions. It is a simple diagonal preconditioner.

Deep networks cannot pre-scale all internal activations by hand; BatchNorm and LayerNorm provide related internal normalization mechanisms. Their optimization effects are studied in ES 667.

Checkpoint: units and curvature

Q QUESTION

You re-measure the "area" feature in centimeters² instead of m² — every value grows $\times 10^4$. In the least-squares loss, the curvature along that weight's axis becomes…

A. unchanged — units can't matter

B. $\times 10^4$

C. $\times 10^8$

D. $\div 10^4$

Answer: units and curvature

A ANSWER

Correct: C — $\times 10^8$

Why: Curvature along w_j is $\mathrm{mean}(x_j^2)$ — the feature scale enters **squared**: $(10^4)^2 = 10^8$. One careless unit choice, eight orders of magnitude of conditioning damage (and the safe η shrinks by 10^8 to match).

Same lesson from the other side: the **optimum weight** shrinks by 10^4 — the minimum didn't move in "price space", but the bowl got squashed 10^8-fold along that axis.

Momentum: memory against the ravine

A marble doesn’t zig-zag

Roll a heavy marble down the ravine instead of teleporting a point:

- The gradient acts as a **force**; the marble accumulates **velocity**.
- Across the valley the force flips direction every instant — the pushes **cancel** in the velocity.
- Along the valley the force always agrees — the pushes **add up**, and the marble accelerates.

Same information (∇f), one new state variable (velocity) — and the ravine’s two personalities get opposite treatment. This is the heavy-ball method.

The update rule: two lines now

$$v_{t+1} = \beta\, v_t + \nabla f(\theta_t), \qquad \theta_{t+1} = \theta_t - \eta\, v_{t+1}$$

```
v = 0
for t in range(T):
    g = grad(L, theta)
    v = beta * v + g            # remember where we've been going
    theta = theta - lr * v      # step along the memory, not the gradient
```

- $\beta \in [0, 1)$ is the **momentum coefficient** — how much velocity survives each step. Default: 0.9.

Unroll it: a moving average of gradients

Feed the recursion into itself:

$$v_t = g_t + \beta g_{t-1} + \beta^2 g_{t-2} + \beta^3 g_{t-3} + \cdots$$

- A **geometrically-weighted memory** of every gradient so far: recent ones loud, old ones fading by β each step.
- GD is the $\beta = 0$ special case: memory of exactly one gradient.

You have met this object: it is a geometric series in disguise — the same $1 + \beta + \beta^2 + \cdots = 1/(1-\beta)$ that priced L2′s float gaps and L15′s discounted pseudo-counts.

Numbers: cancel vs accumulate

$\beta = 0.9$. Compare the two axes of the ravine, in steady state:

Along the valley — gradients keep agreeing, say $g = -0.4$ every step:

$$v \to -0.4(1 + \beta + \beta^2 + \cdots) = \frac{-0.4}{1 - 0.9} = -4 \quad \text{— amplified } \times 10$$

Across the valley — gradients alternate $+2, -2, +2, \ldots$:

$$v \to 2(1 - \beta + \beta^2 - \cdots) = \frac{2}{1 + 0.9} \approx 1.05 \quad \text{— damped to half}$$

Momentum is a direction filter: consistent signals gain $1/(1-\beta)$, flip-flopping signals shrink to $1/(1+\beta)$.

How much memory? $1/(1-\beta)$ steps

β	**effective memory**	**character**
0.5	≈ 2 steps	barely more than GD
0.9	≈ 10 steps	the workhorse default
0.99	≈ 100 steps	a supertanker — smooth, slow to turn

- Along consistent directions the **effective** step is $\approx \eta/(1-\beta)$ — budget η accordingly (that's why the momentum run below uses a smaller η).

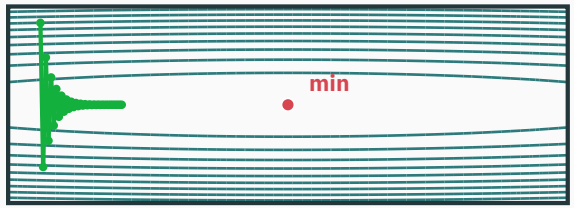
Too much memory overshoots: the marble sails past the valley's end and **rings** back and forth. If the loss oscillates on a slow timescale, suspect β, not η.

The ravine, rerun

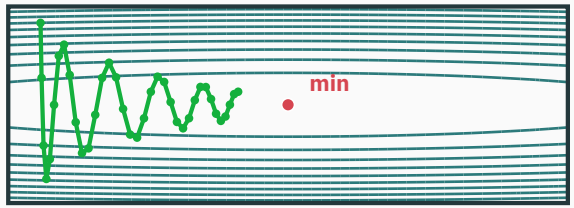
V VISUAL

Same bowl ($\kappa = 160$), same start, same 36 steps:

plain GD · $\eta = 0.55$



momentum · $\eta = 0.21, \beta = 0.9$



the cross-valley wobble is filtered out; the along-valley coordinate — GD's crawl — actually **arrives**

Momentum doesn't shrink κ — it changes how much κ hurts.

Interactive: the momentum map

I INTERACTIVE

INTERACTIVE

↗ distill.pub/2017/momentum/

Distill, **“Why Momentum Really Works”** — this week’s assigned reading, and the most beautiful interactive article in optimization. Drag η and β across the 2-D stability map; watch the ravine’s per-axis rates respond exactly as today’s algebra predicts.

As you read, test yourself: the article’s “effective step size” $\eta/(1-\beta)$, its condition-number plots, its ringing regime — every one of them appeared on a slide today.

Summary & what’s next

The whole lecture in one table

symptom (loss curve)	**diagnosis**	**medicine**
barely-tilted straight decay	$\eta a_{\min} \ll 1$: crawling	raise η toward the wall
loss rises geometrically	$\eta > 2/a_{\max}$: past the wall	cut η (the classic $\div 10$)
fast drop, then endless crawl	$\kappa \gg 1$: ravine	**standardize features**; add **momentum**
slow large-period ringing	β too high: supertanker	lower β before touching η

Every row follows from $x_{t+1} = (1 - \eta a)x_t$ in a different region of (η, a, κ) space.

Lecture 17 — summary

- **GD derived** ⭐: use L7′s local line for one step — $\theta_{t+1} = \theta_t - \eta \nabla f$, with first-order decrease $\eta \|\nabla f\|^2$ (direction: L8′s Cauchy–Schwarz).
- **Learning-rate regimes**: slow convergence / fast convergence / oscillation / divergence; the boundary is $\eta = 2/a$.
- **Quadratic local model** ⭐: iterates are $(1 - \eta a)^t x_0$; per-axis factors on positive-definite bowls; best 1-D rate at $\eta = 1/a$.
- **Conditioning**: $\kappa = a_{\max}/a_{\min}$; the best contraction factor is $(\kappa - 1)/(\kappa + 1)$, so the required steps scale with κ.
- **Preconditioning**: feature scale enters the Hessian quadratically; standardizing removes scale imbalance but not feature correlation.
- **Momentum**: $v = \beta v + g$; consistent directions $\times\, 1/(1 - \beta)$, flip-flops damped — ravines tamed, not cured.

Lecture 17 — summary

Read before L18: MML §7.1 (gradient descent) · Distill, **Why Momentum Really Works** (assigned). **T8** this week: GD convergence on quadratics, condition numbers, and the optimizer race.

Practice problems

Try on paper; verify in the T8 notebook.

1. For $f(x) = 2x^2$ with $\eta = 0.4$: compute the contraction factor and the first three iterates from $x_0 = 1$. Converging?
2. On $f(x) = \frac{5}{2}x^2$: find every η that converges, the η that lands in one step, and what happens at exactly $\eta = 0.4$.
3. On $f(x, y) = \frac{1}{2}(x^2 + 16y^2)$ with the best single η: per-axis factors, and how many steps to shrink the slow-axis error $1000\times$?
4. Two uncorrelated features have standard deviations 1 and 50 (means 0). Estimate κ before and after standardization. What changes if they are nearly collinear?
5. Momentum with $\beta = 0.95$: the effective memory length, and the limiting $|v|$ under alternating gradients $\pm g$.

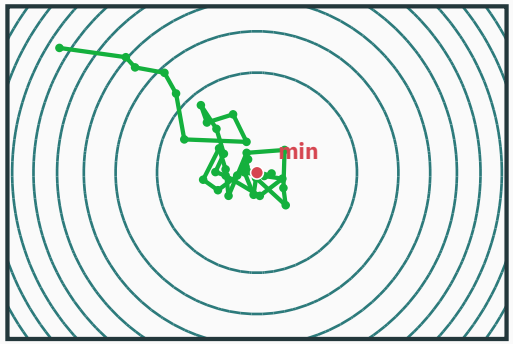
Practice problems

6. Reproduce the opening pair in NumPy ($a = 0.02, 3.2$, $\eta = 0.8$, 30 steps). Find the largest η that converges on both and the steps the flatter problem needs for a $100\times$ error reduction.

⭐⭐⭐ SGD exists

★ OPTIONAL

A finite-sum loss is $\mathcal{L}(\theta) = \frac{1}{n}\sum_i \ell_{i(\theta)}$. If index i is sampled uniformly, then $\mathbb{E}_i\left[\nabla \ell_{i(\theta)}\right] = \nabla \mathcal{L}(\theta)$. One-example SGD uses this unbiased but noisy estimate and usually costs much less per update than a full-data gradient.



Minibatches, learning-rate schedules, and the adaptive per-coordinate rates of RMSProp and Adam belong to ES 335 and ES 667. The contraction-factor analysis here is their quadratic baseline.

The learning rate sets the step length used with a local linear approximation.

Next: **Newton & Gauss–Newton** — what if you trusted a **quadratic** model instead?